higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

PROCESS CONTROL
NQF LEVEL 3

(6050403)

13 March 2019 (X-Paper)
09:00–12:00

This question paper consists of 5 pages.

NCV

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

NCV

QUESTION 1

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1. Soft alarms are found within a DCS or PLC.

1.2 A low-water interlock allows fuel to shut off upon a low-water signal.

1.3 Oscillation is a movement up and down or back and forward.

1.4 Deviation shows how much a measured value differs from a fixed or standard value.

1.5 An inclined-leg manometer is used to measure very high pressures.

1.6 Mercury is used as the liquid in a manometer because of its high density and low vapour pressure.

1.7 Direct flowmeters are generally used to measure and indicate total flow.

1.8 An open-loop control system does not need a human operator to take corrective measures.

1.9 A purge interlock allows fuel to be admitted into an unclean furnace.

1.10 When utilising feedback control a correction of a disturbance can be made before the disturbance is known.

(10 × 1) **[10]**

QUESTION 2

Give ONE term for each of the following descriptions by choosing a term from the list below. Write only the term next to the question number (2.1–2.5) in the ANSWER BOOK.

primary device; recording; error; alarm; controller

2.1 Instrument used to give early warning of failures that may happen in a process

2.2 Instrument used to make sure that a product complies with quality requirements

2.3 Instrument used to measure the value of a process variable, for example pressure level and flow, to control a process.

2.4 Measured values indicated on a gauge and stored on a computer to be used when needed

2.5 Output of a controller fed to a correcting unit

(5 × 1) **[5]**

Copyright reserved

QUESTION 3

Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–H) next to the question number (3.1–3.5) in the ANSWER BOOK.

COLUMN A		COLUMN B	
3.1	Fluid enters the inlet and fills the space between the lobes and case causing them to rotate as fluid moves through the meter	A	oscillating piston
3.2	At each cycle the piston completes, a known volume of fluid leaves the meter	B	oval gear meter
3.3	Vanes are rigidly attached to cam rollers and can slide freely through the rotor	C	nutating disc
3.4	Meter using Faraday's law for electromagnetic inductance	D	orifice plate
3.5	Device creating differential pressure	E	magnetic flowmeter
		F	reciprocating piston
		G	turbine flowmeter
		H	rotary vane meter

(5 × 1) **[5]**

QUESTION 4

4.1 What is the purpose of a control chart? (2)

4.2 What is meant by the term *upper control limit*? (2)

4.3 What is the operating range of a pneumatic signal? (2)

4.4 Give the purpose of performing a zero check on an instrument. (2)

4.5 Give the purpose of calibrating an instrument. (2)

[10]

Copyright reserved

QUESTION 5

5.1	Explain the working principles of a sight glass.		(6)
5.2	Explain the operating principles of a solid-rod thermostat.		(6)
5.3	Explain, with the aid of a diagram, the operating principles of a separator drum.		(13)
5.4	Explain the working principles of a bimetallic thermometer.		(10)
5.5	State THREE maintenance requirements for instruments.	(3 × 2)	(6)
			[41]

QUESTION 6

6.1	Explain the working principles of a rotating-paddle level-measurement instrument.		(3)
6.2	State the SIX steps to follow in systematic problem analysis.		(6)
6.3	Give FOUR advantages and FOUR disadvantages of an orifice plate.	(4 + 4)	(8)
6.4	Give FOUR advantages and FOUR disadvantages of a venturi tube.	(4 + 4)	(8)
6.5	Give the purpose of a control loop.		(4)
			[29]

TOTAL: 100

Copyright reserved